Effects of Velocity-Based Resistance Training on Young Soccer Players of Different Ages

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Abstract

González-Badillo, JJ, Pareja-Blanco, F, Rodríguez-Rosell, D, Abad-Herencia, JL, del Ojo-López, JJ, and Sánchez-Medina, L. Effects of velocity-based resistance training on young soccer players of different ages. *J Strength Cond Res* 29(5): 1329–1338, 2015—This study aimed to analyze the effect of velocity-based resistance training (RT) with moderate loads and few repetitions per set combined with jumps and sprints on physical performance in young soccer players of different ages. A total of 44 elite youth soccer players belonging to 3 teams participated in this study: an under-16 team (U16, $n = 17$) and an under-18 team (U18, $n = 16$) performed maximal velocity RT program for 26 weeks in addition to typical soccer training, whereas an under-21 team (U21, $n = 11$) did not perform RT. Before and after the training program, all players performed 20-m running sprint (T20), countermovement jump (CMJ), a progressive isoinertial loading test in squat to determine the load that elicited a $\sim$1 m·s^{-1} velocity (V1LOAD) and an incremental field test to determine maximal aerobic speed (MAS). U16 showed significantly ($p = 0.000$) greater gains in V1LOAD than U18 and U21 (100/0/0%). Only U16 showed significantly ($p = 0.01$) greater gains than U21 (99/1/0%) in CMJ height. U18 obtained a likely better effect on CMJ performance than U21 (89/10/1%). The beneficial effects on T20 between groups were unclear. U16 showed a likely better effect on MAS than U21 (80/17/3%), whereas the rest of comparisons were unclear. The changes in CMJ correlated with the changes in T20 ($r = -0.49$) and V1LOAD ($r = 0.40$). In conclusion, velocity-based RT with moderate load and few repetitions per set seems to be an adequate method to improve physical performance in young soccer players.

Key Words strength, team sport, physical performance, conditioning

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Introduction

During an elite-level soccer match, players run 10–12 km at a moderate average intensity (32). In the aerobic context in which the match is developed, the most crucial events are represented by high-intensity work, as the majority of the goals are preceded by a lineal sprint, vertical jump, or change of direction of the scoring or the assisting player (11). Such actions require high strength and power generation by the muscles of the lower limbs (27). In this regard, it has been reported that jump height, 10-m sprint, and 30-m sprint performances are correlated ($r =$ 0.78, 0.94, and 0.71, respectively) with maximal muscular strength in professional soccer players (39). The positive effects of resistance training (RT) on strength, jumping, and sprinting abilities in adult soccer players have been widely studied (17,18). Nevertheless, the effect of RT on endurance in soccer players has received less scientific attention.

Resistance training has shown to have beneficial effects on the muscular power and motor skill performance of adolescent athletes (16). However, little information is available in the literature concerning young soccer players. Most of the studies conducted with young soccer players used plyometric training (4,6,19,27,38). However, studies that included a RT program generally used repetitions to muscular failure in each set or high loads (70–95% of one-repetition maximum [1RM]), even though these studies were performed with athletes with no previous RT experience. (5,7,21,25,26,28,40). Moreover, heavy-load training (3–6RM) is rarely possible during the soccer competitive season because this type of training produces an excessive fatigue that does not allow players to undertake effective ball practice after this form of concurrent training (10). Other studies (15,23,29) used lower intensities, although 2 of them (15,29) expressed relative intensity as percentage of body mass, which may hinder the interpretation of the stress produced by the RT performed.

The assessments of 1RM or XRM have been common methods to analyze strength performance (5,7,19,21,25,26,28,31,37). However, this type of test requires a great experience from the subjects, and it might suppose an unnecessary risk and stress for the athletes. The introduction of new technology (linear position transducers, rotary encoders, accelerometers, etc.) now enables the direct

Table 1. Age, height, and body mass of the 3 groups, mean ($\pm SD$).*

	U16	U18	U21
Age (yr)	14.9 ± 0.3	17.8 ± 0.4	19.2 ± 1.2
Height (cm)	175.5 ± 5.6	176.1 ± 6.2	178.1 ± 6.7
Body mass (kg)	67.7 ± 9.2	73.7 ± 9.2	75.5 ± 4.4

*U16 = under-16 group ($n = 17$); U18 = under-18 group ($n = 16$); U21 = under-21 group ($n = 11$).

measurement of many kinematic and kinetic variables that can be used to assess the effects of RT on performance. Several studies have used power (24,31) or velocity as reference to prescribe and monitor RT (13,30). However, great variability has been found in the relative loads that produce maximal power (20–80% 1RM), depending on the method used to measure maximal power (8,9,35). Finally, it has been recently shown that movement velocity has a very close relationship with the %1RM (14,35); that is, if 2 subjects lift a given load at the same absolute velocity, we could determine that both subjects are working at the same %1RM. An interesting study by López-Segovia et al. (23) used velocity as reference to prescribe RT in youth soccer player during 16 weeks. In this study, relative loads ranging from 45% 1RM (i.e., 1.20 $m \cdot s^{-1}$) to 70% 1RM (i.e., 0.80 $m \cdot s^{-1}$) were used, in addition to a low number of repetitions with respect to the maximum number that can be completed with these loads (4–8 repetitions) combined with vertical jumps and sprints. López-Segovia et al. (23) suggested that a velocity-based RT, in which were used loads lesser than 70% 1RM, might produce enhancement in strength performance in soccer players without the need to perform maximum repetitions.

Despite the perceived and demonstrated importance of strength and speed in soccer (33), to the best of our knowledge, no study has addressed the issue of speed and strength development and the relationships between the changes produced in different fitness indicators by RT in youth soccer players of different ages. Furthermore, it seems that more studies are necessary that analyze the effects of RT with moderate loads on different physical abilities in youth soccer players. For these reasons, the main aim of this study was to analyze the effect of adding adding velocity-based RT to the typically technical-tactical soccer training. This in-season RT program used moderate loads and a low number of repetitions per set performed at maximal intended velocity, and it was combined with vertical jumps and sprints. The effect of RT on lower-body strength, jumping height, acceleration and endurance capacity was compared in young soccer players of different ages.

Methods

Experimental Approach to the Problem

A quasi-experimental design was used. Three groups of soccer players from the same club participated in this study: an under-16 team (U16) and an under-18 team (U18) performed 2 RT sessions per week, and an under-21 team (U21) performed only typical soccer training. During the competitive season, subjects trained 4 times and played 1 official soccer game per week. All players undertook a battery of tests for the evaluation of their performance at the end of the preseason and a posttest after 26 weeks of training. The battery of tests was performed in 2 sessions. The first testing session consisted of (a) 20-m all-out running sprints (T20), (b) countermovement jumps (CMJ), and (c) a progressive isoinertial loading test in the full squat exercise. In the second testing session, players completed an incremental field running test to determine maximal aerobic speed (MAS). All tests were carried out at least 48 hours after the most recent game and took place at a neuromuscular research laboratory, except the MAS test that was performed in an athletics track, under the direct supervision of the investigators at the same time of the day (±1 hour) for each subject and under constant environmental conditions (~20° C, ~60% humidity). Between the initial tests (pretest) and the final tests (posttest), there were 26 weeks of training from the beginning of September to the end of March. During the 2 weeks preceding this study, 4 preliminary familiarization sessions were undertaken with the purpose of emphasizing proper execution technique in the full squat and CMJ. The tests executed for the assessment of performance are explained in detail below.

Subjects

At the beginning of the study, a total of 58 elite youth soccer players (age range, 14–21 years) participated in this investigation. The soccer players were members of the development program of the same first-division professional soccer club in Spain. According to the soccer federation rules, players are matched and compete by chronological age rather than biological maturation. Thus, in this study, players were pooled by age group, exactly the way they are matched in training and competition. All soccer players who participated in this study had no experience in strength training. During the study, 14 players quit because of injury or illness not connected with the intervention training regimes. Therefore, at the end of the study, the remaining players were a total of 44 participants (U16, $n = 17$; U18, $n = 16$; U21, $n = 11$). The mean ± SD age, height, and body mass of the 3 groups are displayed in Table 1. Written consent was obtained from the participants and the participants' parents/guardians if the player was younger than 18 years old, after being thoroughly informed of the purpose and potential risks of the study. The investigation was conducted in accordance

TABLE 2. Resistance training program.*†

Weeks	Pretest	1	2	3	4	5	6	7
SQ (%V1LOAD)		2 × 8 (80%)	3 × 8 (80%)	3 × 6 (90%)	3 × 8 (90%)	2 × 6 (100%)	3 × 6 (100%)	Control test
CMJ_L (%load-20 cm)		3 × 6 (40%)	3 × 6 (50%)	3 × 6 (60%)	3 × 6 (60%)	3 × 6 (70%)	3 × 6 (70%)	
JB		3 × 5	3 × 5	3 × 5	3 × 5	3 × 5	3 × 5	
Sled towing								
SPTJ (steps)		3 × (2 × 6)	3 × (2 × 6)	3 × (2 × 6)	3 × (2 × 8)	3 × (2 × 8)	3 × (2 × 8)	
COD		3 × 10″	3 × 10″	4 × 10″	4 × 10″	5 × 10″	3 × 10″	
Sprint		3 × 20 m	4 × 20 m	3 × 20 m	4 × 20 m	4 × 20 m	3 × 20 m	

Weeks	8	9	10	11	12	13	14	15
SQ (%V1LOAD)	3 × 6 (85%)	3 × 6 (85%)	3 × 6 (95%)	3 × 6 (95%)	3 × 6 (95%)	3 × 4 (105%)	3 × 4 (105%)	Control test
CMJ_L (%load-20 cm)	3 × 4 (40%)	3 × 4 (50%)	3 × 4 (60%)	3 × 4 (60%)	3 × 4 (70%)	3 × 4 (70%)	3 × 4 (70%)	
JB	3 × 5	3 × 5	3 × 5	3 × 5	3 × 5	3 × 5	3 × 5	
Sled towing	4 × 20 m	4 × 20 m	4 × 20 m	4 × 20 m	4 × 20 m	4 × 20 m	4 × 20 m	
SPTJ (steps)	3 × (2 × 8)	3 × (2 × 8)	3 × (2 × 8)	3 × (2 × 8)	3 × (2 × 8)	3 × (2 × 8)	3 × (2 × 8)	
COD	4 × 10″	4 × 10″	4 × 10″	4 × 10″	4 × 10″	4 × 10″	4 × 10″	
Sprint	3 × 20 m	3 × 20 m	3 × 20 m	3 × 20 m	3 × 20 m	3 × 20 m	3 × 20 m	

Weeks	16	17	18	19	20	21	22	23	24	25	26	Posttest
SQ (% V1LOAD)	2 × 8 (90%)	3 × 8 (90%)	3 × 6 (95%)	3 × 6 (95%)	4 × 6 (95%)	4 × 6 (95%)	3 × 6 (100%)	4 × 6 (100%)	3 × 4 (105%)	3 × 4 (105%)	3 × 4 (105%)	
CMJ_L (%load-20 cm)	3 × 5 (60%)	4 × 5 (60%)	4 × 5 (60%)	4 × 5 (60%)								
HJ	3 × 4 × 2	4 × 4 × 2	4 × 4 × 2	2 × 4 × 3	2 × 4 × 3	3 × 4 × 3	4 × 4 × 3	2 × 4 × 3	3 × 5 × 3	3 × 5 × 3	3 × 5 × 3	
Sled towing					4 × 25 m	4 × 25 m	5 × 25 m	5 × 25 m	5 × 25 m	4 × 20 m	4 × 20 m	
SPTJ (steps)	3 × (2 × 8)	4 × (2 × 8)	4 × (2 × 8)	4 × (2 × 8)	4 × (2 × 8)	4 × (2 × 8)	4 × (2 × 8)	4 × (2 × 8)	4 × (2 × 8)	4 × (2 × 8)	3 × (2 × 8)	
COD	3 × 10″	4 × 10″	5 × 10″	5 × 10″	5 × 10″	5 × 10″	5 × 10″	5 × 10″	5 × 10″	4 × 10″	3 × 10″	
Sprint	4 × 20 m	4 × 20 m	5 × 20 m	5 × 20 m	3 × 20 m	3 × 20 m	3 × 20 m	3 × 20 m	3 × 20 m	3 × 20 m	3 × 20 m	

*SQ = squat; CMJ_L = countermovement jump with load; HJ = hurdle jumps; %V1LOAD = percentage of the load that elicited ~1 $m \cdot s^{-1}$ in the squat test; %load-20 cm = percent of the load with which the players jumped ~20 cm in the loaded countermovement jump exercise; JB = jump to box; SPTJ = step phase triple jumps; COD = acceleration with changes of direction sprint.

†The squat exercise was performed twice per week, and the rest of exercises were performed once a week.

TABLE 3. Changes in selected neuromuscular performance variables from T1 to T2 for each group, mean (±*SD*).*

			Changes observed for T2 vs. T1		
	T1	T2	*p* value within groups	Standarized (Cohen) differences (90% CI)	Percent changes of better/trivial/worse effect
CMJ-U16 (cm)	35.4 ± 3.9	39.1 ± 4.9	0.000	0.91 (0.70 to 1.11)	100/0/0 most likely
CMJ-U18 (cm)	38.4 ± 3.0	41.3 ± 4.5	0.000	0.90 (0.45 to 1.35)	99/1/0 very likely
CMJ-U21 (cm)	37.1 ± 3.7	38.1 ± 3.5	0.36	0.18 (−0.14 to 0.50)	45/52/3 possibly
T20-U16 (s)	2.99 ± 0.10	2.97 ± 0.09	0.14	0.23 (0.08 to 0.38)	62/38/0 possibly
T20-U18 (s)	2.96 ± 0.10	2.92 ± 0.10	0.02	0.37 (0.11 to 0.63)	87/13/0 likely
T20-U21 (s)	2.97 ± 0.09	2.96 ± 0.10	0.36	0.18 (−0.32 to 0.68)	47/43/10 unclear
V1LOAD-U16 (kg)	41.7 ± 9.3	69.9 ± 12.5	0.000	2.86 (2.60 to 3.12)	100/0/0 most likely
V1LOAD-U18 (kg)	51.6 ± 10.7	66.6 ± 10.1	0.000	1.31 (1.10 to 1.53)	100/0/0 most likely
V1LOAD-U21 (kg)	53.1 ± 4.9	65.9 ± 2.2	0.000	2.38 (1.96 to 2.79)	100/0/0 most likely
MAS-U16 ($km \cdot h^{-1}$)	15.9 ± 0.7	16.2 ± 0.8	0.02	0.52 (0.20 to 0.84)	95/5/0 likely
MAS-U18 ($km \cdot h^{-1}$)	15.8 ± 1.0	16.0 ± 0.8	0.12	0.24 (−0.03 to 0.51)	60/40/0 possibly
MAS-U21 ($km \cdot h^{-1}$)	15.9 ± 0.7	15.9 ± 0.8	0.91	0.03 (−0.48 to 0.54)	28/50/22 unclear

*CI = confidence interval; CMJ = countermovement jump height; U16 = under-16 group ($n = 17$); U18 = under-18 group ($n = 16$); U21 = under-21 group ($n = 11$); T20 = 20-m sprint time; V1LOAD = the load that elicited ~1.00 $m \cdot s^{-1}$ in the full-squat exercise; MAS = maximal aerobic speed test (University of Montreal Track Test).

with the Declaration of Helsinki and approved by the Ethics Committee of Pablo de Olavide University.

Procedures

Running Sprints. Two 20-m sprints, separated by a 3-minute rest, were performed in an indoor running track. Photocell timing gates (Polifemo Radio Light; Microgate, Bolzano, Italy) were placed at 0 and 20 m. A standing start with the leadoff foot placed 1 m behind the first timing gate was used. All participants were required to give an all-out maximal effort in each sprint, and the best of both trials was kept for analysis. Performance feedback was given for each player after the execution of each sprint. The same warm-up protocol that incorporated several sets of progressively faster 30-m running accelerations was followed in the pre- and posttests.

Vertical Jump. Jump height was calculated at the nearest 0.1 cm from flight time measured with an infrared timing system (Optojump; Microgate). The displacement of the center of gravity during the flight was estimated as jumping height (h), which was calculated using the recorded flight time as follows (3): $h = (g \times ft^2) \cdot 8^{-1}$, where "$g$" is the acceleration of gravity (9.81 $m \cdot s^{-2}$) and "ft" is the flight time. Because the takeoff and landing position can affect the jump flight, strict instructions were addressed to all participants to keep their legs straight during the flight time of the jump. The player starts from an upright standing position, makes a downward movement until approximating a knee angle of 90°, and subsequently begins to push off at a maximal intended velocity. All participants completed 5 maximal CMJs with their hands on their hips separated by 1-minute rest. All players received feedback about their performance between trials. The highest and lowest values were discarded, and the resulting average value was kept for analysis.

Isoinertial Progressive Loading Test. The assessment consisted of an isoinertial test with increasing loads using the full-squat exercise performed in a Smith machine (Multipower Fitness Line; Peroga, Murcia, Spain). The squat was performed with subjects starting from the upright position with the knees and hips fully extended, stance approximately shoulder-width apart, and the barbell resting across the back at the level of the acromion. Each participant descended in a continuous motion until the top of the thighs got below the horizontal plane, then immediately reversed motion, and ascended back to the upright position. Feedback based on eccentric distance traveled and concentric velocity was provided. This was accomplished by using a linear velocity transducer (described later in detail) that registered the kinematics of every repetition and whose software provided visual and auditory feedback in real-time. Unlike the eccentric phase that was performed at a controlled mean velocity (i.e., 0.50–0.65 $m \cdot s^{-1}$), athletes were required to always execute the concentric phase of each repetition at maximal intended velocity, that is, explosively. Strong verbal encouragement and velocity feedback in every repetition was provided to motivate the participants to give a maximal effort. Initial load was set at 30 kg and was progressively increased in 10-kg increments. Players performed 3 repetitions with each load. Only the best repetition at each load,

Table 4. Changes in selected neuromuscular performance variables from T1 to T2 between groups.*†

	Changes observed for T2 vs. T1		
	p value between groups	Standarized (Cohen) differences (90% CI)	Percent changes of better/trivial/ worse effect
CMJ			
U16 vs. U18	0.69	0.31 (−0.07 to 0.69)	69/29/2 possibly
U16 vs. U21	0.01	0.77 (0.40 to 1.13)	99/1/0 very likely
U18 vs. U21	0.12	0.56 (0.07 to 1.04)	89/10/1 likely
T20			
U16 vs. U18	0.87	−0.14 (−0.43 to 0.15)	3/60/37 possibly trivial
U16 vs. U21	0.99	0.06 (−0.44 to 0.56)	31/50/19 unclear
U18 vs. U21	0.77	0.21 (−0.33 to 0.75)	51/39/10 unclear
V1LOAD			
U16 vs. U18	0.00	1.13 (0.84 to 1.43)	100/0/0 almost certainly
U16 vs. U21	0.00	1.49 (1.18 to 1.80)	100/0/0 almost certainly
U18 vs. U21	0.67	0.24 (−0.10 to 0.58)	57/41/2 possibly
MAS			
U16 vs. U18	0.90	0.15 (−0.25 to 0.55)	42/51/7 unclear
U16 vs. U21	0.38	0.50 (−0.10 to 1.09)	80/17/3 likely
U18 vs. U21	0.73	0.24 (−0.25 to 0.74)	56/37/7 unclear

*CI = confidence interval; CMJ, countermovement jump height; U16 = under-16 group ($n = 17$); U18 = under-18 group ($n = 16$); U21 = under-21 group ($n = 11$); T20 = 20-m sprint time; V1LOAD = the load that elicited ~1.00 $m \cdot s^{-1}$ in the full-squat exercise; MAS = maximal aerobic speed test (University of Montreal Track Test).

†For clarity, all differences are presented as improvements for the first group compared with the second group (i.e., U16 vs. U18), so that negative and positive differences are in the same direction.

according to the criteria of fastest mean propulsive velocity (36), was considered for subsequent analysis. Four-minute rests were taken between sets. The test ended for each player when the mean propulsive velocity was ~1.00 $m \cdot s^{-1}$ (range: 0.95–1.05 $m \cdot s^{-1}$). This value was chosen for several reasons: (a) the maximal load used in squat exercise during RT was the load that elicited ~1.00 $m \cdot s^{-1}$ (V1LOAD), which represents ~56% 1RM (1), thereby, providing enough information for training prescription; (b) larger weights may predispose to a higher risk of ventral flexion of the lumbar spine while squatting; (c) this load has already been used in a previous study as reference to prescribe the RT (23). A total of 5.3 ± 1.8 increasing loads were used for each player. The warm-up consisted of 5 minutes of joint mobilization exercises, followed by 2 sets of 8 and 6 repetitions (3-minute rest) with loads of 20 and 30 kg, respectively. The exact same warm-up and progression of absolute loads were repeated in the post-test by each participant. V1LOAD was used to assess strength performance. A dynamic measurement system (T-Force System; Ergotech, Murcia, Spain) automatically calculated the relevant kinematic parameters of every repetition, provided auditory and visual velocity feedback in real-time and stored data on disk for analysis. This system consists of a linear velocity transducer interfaced to a personal computer by means of a 14-bit resolution analog-to-digital data acquisition board and custom software. Instantaneous velocity was sampled at 1,000 Hz and subsequently smoothed using a fourth order low-pass Butterworth filter with no phase shift and a cutoff frequency of 10 Hz. Reliability of this system has been recently reported elsewhere (34). The velocity measures used in this study correspond to the mean velocity of the propulsive phase of each repetition. The propulsive phase was defined as that portion of the concentric phase during which the measured acceleration is greater than acceleration due to gravity (i.e., $a > -9.8$ $m \cdot s^{-2}$) (36).

Incremental Field Running Test. A modified version of the University of Montreal Track Test (22) was performed. The test was conducted on a 400-m track with visual markers every 25 m. The initial speed of the test was 8 $km \cdot h^{-1}$. Similar to the original test, this test was performed with a progression of 1 $km \cdot h^{-1}$ every 2 minutes, but the increment of speed was continuous. The athletes followed a speed that was determined by audio cues, and the test ended when the players failed on 2 consecutive occasions to reach the next cone in the required time. Maximal aerobic speed was the speed that corresponded to the last stage completed by the participants according to the established protocol (22).

Resistance Training Program. For groups U16 and U18, this training was complemented with 2 specific strength training sessions in the weight room with durations of 30–45 minutes before the field training. Strength training was carried out using essential exercises such as full squat, and jumps with

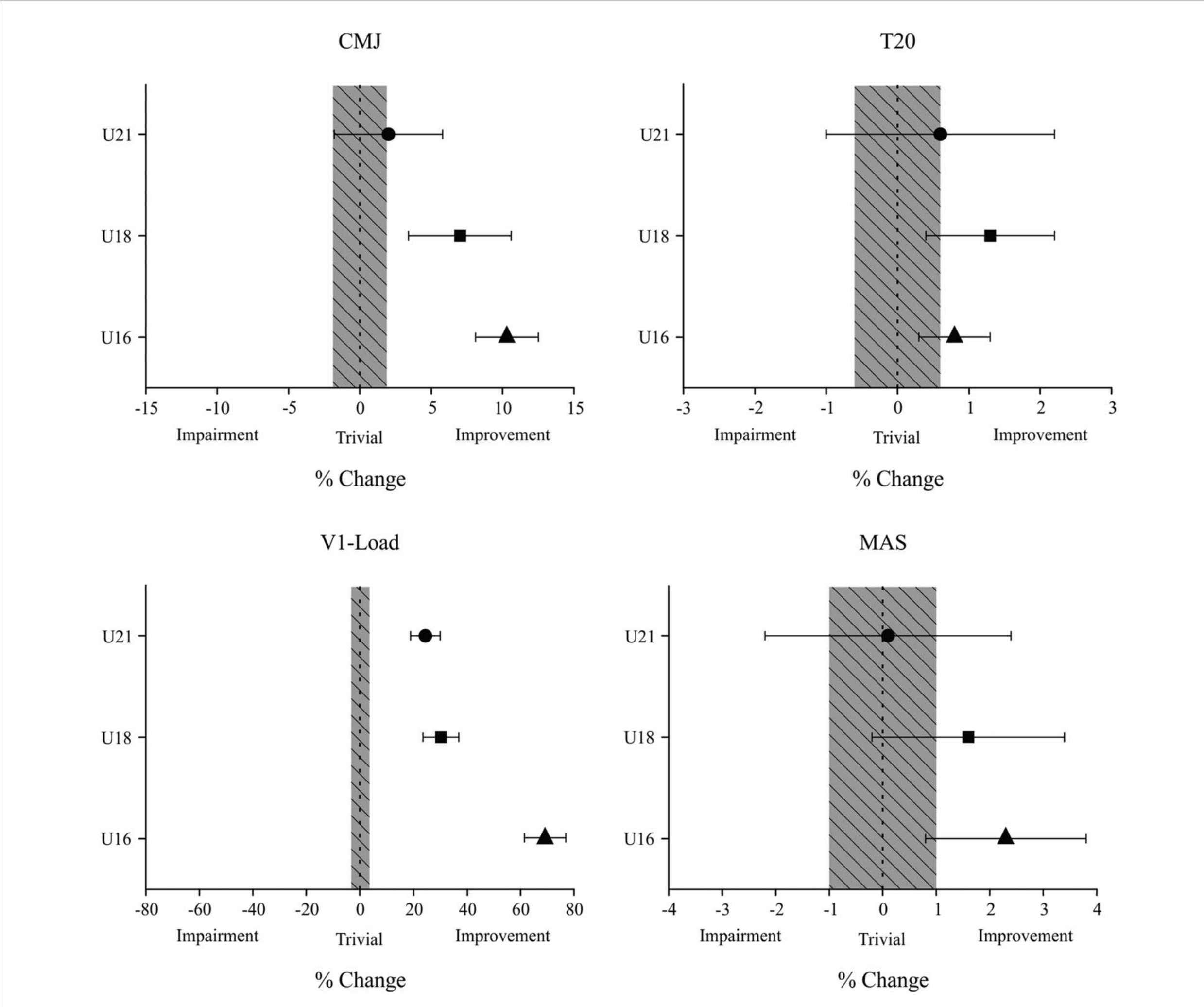

Figure 1. Within-group relative changes for under-16 group (U16), under-18 group (U18), and under-21 group (U21) in countermovement jump (CMJ) height, 20-m sprint time (T20), the load that elicited ~1.00 m·s^{-1} in the full-squat exercise (V1LOAD), and maximal aerobic speed test (University of Montreal Track Test). Bars indicate uncertainty in the true mean changes with 90% confidence intervals. The trivial area was calculated from the smallest worthwhile change.

(CMJ_L). Furthermore, jumps to box and hurdle jumps were performed. Box and hurdle heights were adjusted for each subject. Table 2 shows in detail the exercises, number of sets and repetitions, and the exercise intensities. The relative loads used by each player were assigned according to the V1LOAD for full squat and the load with which the players were able to jump ~20 cm for the CMJ_L. In the 7th and 15th weeks, V1LOAD and the load with which each subject jumped ~20 cm were recalculated for the sessions of the following weeks. Approximately 3-minute rest periods were allowed between each set and each exercise. The players were instructed to perform all exercises at maximal intended velocity. Two trained researchers supervised each workout session and recorded the compliance and individual workout data during each training session. Strength training was completed in the field by the following exercises: 10-second accelerations with changes of direction while carrying a weight disc of 5–10 kg held against the chest; sets of 6–8 executions of the step phase of the triple jump; and running sprints of 20–25 m with or without resited sled towing (5–10 kg) depending on the session.

Statistical Analyses

The values are reported as mean ± *SD*. Statistical significance was established at the level of $p \leq 0.05$. Test-retest absolute reliability was measured by the standard error of the mean (SEM), which was expressed in relative terms through coefficient of variation (CV), whereas relative reliability was assessed by the intraclass correlation coefficients (ICC, 95% confidence interval [CI]) calculated with the 1-way

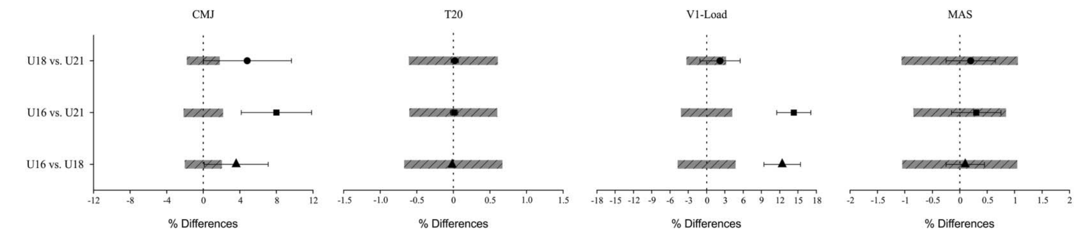

Figure 2. Mean differences in the relative changes between groups for countermovement jump height (CMJ), 20-m sprint time (T20), the load that elicited ~1.00 $m \cdot s^{-1}$ in the full-squat exercise (V1LOAD), and maximal aerobic speed test (University of Montreal Track Test). Bars indicate uncertainty in the true mean changes with 90% confidence intervals. The trivial area was calculated from the smallest worthwhile change. For clarity, all differences are presented as improvements in the first group compared with the second group (i.e., U16 vs. U18), so that negative and positive differences are in the same direction.

random-effects model. The normality of distribution of the differential scores (post-pre values) was examined with the Kolmogorov-Smirnov test, and the homogeneity of variance assumption was not violated. A 1-way analysis of variance (ANOVA) was conducted to examine between-group differences with Gabriel's post hoc comparisons (U16 vs. U18 vs. U20). A related-sample *t*-test was used to analyze intragroup changes between pre- and posttraining. The statistical analyses were performed using SPSS software version 18.0 (SPSS Inc., Chicago, IL, USA). In addition to this null hypothesis testing, the data were assessed for clinical significance using an approach based on the magnitudes of change (2,20). The effect sizes (ESs) were calculated using Hedge's *g*. A practically worthwhile change was assumed when the difference score was at least 0.2 of the between-subject *SD*. Quantitative chances of beneficial/better or detrimental/worse effects were assessed qualitatively as follows: <1%, almost certainly not; 1–5%, very unlikely; 5–25%, unlikely; 25–75%, possible; 75–95%, likely; 95–99%, very likely; and >99%, almost certain. If the chances of having beneficial/better or detrimental/worse were both >5%, the true difference was assessed as unclear (2,20). Pearson's correlation coefficients were calculated to establish the respective relationships between the changes of all measured variables. The magnitude of correlation was assessed with the following thresholds: <0.1, trivial; <0.1–0.3, small; <0.3–0.5, moderate; <0.5–0.7, large; <0.7–0.9, very large; and <0.9–1.0, almost perfect (20).

Results

Compliance with the RT program was 98% of all sessions scheduled for the U16 group and 83% for the U18 group. The mean values from pre- to posttraining and the quantitative and qualitative outcomes resulting from the within-group analysis are reported in Table 3. The results from the between-group analysis are shown in Table 4. Test-retest reliability was measured by CV and ICC (95% CI), which were CV = 0.88% and ICC = 0.96 (0.91–0.98) for T20 and CV = 2.89% and ICC = 0.98 (0.97–0.99) for CMJ.

Isoinertial Strength Assessments

The ANOVA performed in the V1LOAD variable showed significant differences ($F_{(2,43)} = 36.27$, $p = 0.0001$, $\eta^2 = 0.68$) for the 3 groups (Table 4). Gabriel's post hoc test showed significant differences in U16 vs. U18 ($p = 0.0001$, ES = 1.13) and U16 vs. U21 ($p = 0.0001$, ES = 1.49). No significant difference was observed between U18 and U21 ($p = 0.67$, ES = 0.24). Mean differences and 95% CI were 13.3 (8.5 to 18.0), 15.5 (10.2 to 20.7), and 2.2 (−3.2 to 7.6) kg for the first, second, and third comparison, respectively. Practically, worthwhile differences seemed evident supported by the substantial greater increases (100/0/0%) in V1LOAD for U16 than U18 and U21 groups. U18 showed possibly (57/41/2%) greater gains for V1LOAD than U21 (Table 4).

Vertical Jump

The ANOVA performed in CMJ variable revealed significant differences ($F_{(2,43)} = 4.51$, $p = 0.02$, $\eta^2 = 0.17$) for the 3 groups. Gabriel's post hoc test showed significant differences in U16 vs. U21 ($p = 0.01$, ES = 0.77). No significant difference was observed in U16 vs. U18 ($p = 0.69$, ES = 0.31) and U18 vs. U21 ($p = 0.12$, ES = 0.56). Mean differences and 95% CI were 3.0 (0.5 to 5.5), 0.9 (−1.4 to 3.2), and 2.1 (−0.4 to 4.6) cm for the first, second, and third comparison, respectively. Furthermore, U16 ($p = 0.0001$, 100/0/0%) and U18 ($p = 0.003$, 99/1/0%) showed significant improvements for jump performance, whereas U18 ($p = 0.32$, 45/52/3%) did not show significant changes (Table 3).

Sprint Ability

No significant difference in T20 variable was observed between groups ($F_{(2,43)} = 0.42$, $p = 0.66$, $\eta^2 = 0.02$). U16 and U18 showed significant ($p \leq 0.05$) improvements (62/48/0% and 87/13/0%, respectively), whereas the changes in running sprints for U21 were unclear (47/43/10%; Figure 1).

Maximal Aerobic Speed

The ANOVA performed in MAS variable did not reveal significant differences ($F_{(2,43)} = 1.08$, $p = 0.35$, $\eta^2 = 0.04$) for the 3 groups. U16 showed a likely better effect on MAS than U21 (80/17/3%) (Figure 2). Furthermore, U16 showed likely improvements ($p = 0.02$, 95/5/0%) and U18 also seemed to improve possibly in MAS ($p = 0.14$, 60/40/0%), whereas the changes in U21 were unclear ($p = 0.92$, 56/37/7%).

Relationships Between Changes in Physical Performance Indices

When the data from the 3 groups were pooled, the relative decrement of T20 was moderately correlated with the relative improvement in CMJ ($r = -0.49$, $p = 0.001$). In addition, the relative improvement in CMJ was significantly correlated with the relative increase in V1LOAD ($r = 0.40$, $p = 0.02$). A moderate negative relationship was observed between the changes in T20 and MAS ($r = -0.50$, $p = 0.001$); however, changes in CMJ height were not related to changes in MAS ($r = 0.28$, $p = 0.07$). Finally, the relative improvements in V1LOAD did not explain the changes in T20 ($r = 0.03$, $p = 0.88$).

Discussion

To the best of our knowledge, this is the first study that analyzes the effect of a velocity-based RT program with moderate loads and a low number of repetitions per set combined with vertical jumps and sprints on lower-body strength, jumping, acceleration, and endurance capacity in young soccer players of different ages. The main finding of this study was that after 26 weeks of training, the U16 and U18 players equaled or even overcame the U21 players (control group), who did not perform the RT program, in all the variables of physical performance analyzed (leg strength, jumping, sprint, and endurance running). Therefore, the results of this study suggest that only 6 months of RT combined with the typical tactical-technical soccer training might provide the same or even greater gains in physical performance than 5 years (from 15–16 to 20–21 years of age) of only typical soccer training.

Most of the isoinertial studies in the literature with young soccer players have commonly used repetitions to failure or high loads (70–95% 1RM) to improve strength and power ability (5,7,21,25,26,28,37,40). However, RT with heavy loads might induce an excessive fatigue that does not allow players to undertake effective ball practice immediately after this type of RT (10). However, in a previous study by López-Segovia et al. (23), lifting velocity was used as a reference to prescribe RT in youth soccer players. In that study, moderate loads (from 45% 1RM ~1.20 $m \cdot s^{-1}$ to 70% 1RM ~0.80 $m \cdot s^{-1}$) and a low number of repetitions per set were used in the squat exercise, but no significant differences in strength gains were found between the RT and control groups. In the present study, 2 researchers were present supervising each workout session and the players were instructed to lift the load at maximum intended velocity every repetition. Lifting the load at maximal velocity seems to be a key factor to optimize the adaptations induced by RT (30). These aspects might explain the substantial improvements in strength performance, mainly for U16. This finding is remarkable because few repetitions and moderate loads (50–65% 1RM) were used, with exercise sets ending well ahead of reaching failure (Table 2). However, it is likely that part of the gains in U16 occurred because of the fact that this group was in a sensitive phase of strength development, as it has been reported in a previous study (12). In any case, it is a relevant finding that U16 obtained equal or even superior strength performance after 6 months of additional RT than U21 after 5 years of typical soccer training.

With regard to physical performance, U16 and U18 obtained better effects on CMJ height and acceleration capacity than U21. These findings are consistent with previous observations of training-induced specificity caused by explosive strength training in which high-velocity contractions with low loads were used (15). A previous study (15) reported that soccer combined with an explosive-type strength training produced significant enhancements in vertical jump and running sprint performance over 5 m, with no interference with the development of endurance running. Another study with a similar design to our study showed improvements in 20-m sprint performance (2–4%) in young soccer players of different ages (37). However, in the study performed by Sander et al. (37), RT was performed throughout 2 years, and heavy loads and repetitions to failure were used, which may produce greater fatigue and do not allow players to perform the technical/tactical/conditioning soccer training in optimum conditions (10). Therefore, the results of the present study suggest that velocity-based RT produces similar gains that a heavy RT performing repetitions to XRM. This fact is important for conditioning in sports that require the improvement of physical performance without producing excessive fatigue that could interfere with the development of other components (technical, tactical or recovery aspects) of training.

Similar to previous studies (15,40), the RT performed in this study did not produce any potential negative effect on aerobic endurance. In fact, it even produced a substantial improvement in the U16 group. This might have been possible because of the fact that this type of RT stresses the neural system but does not place high metabolic demands (34). Thus, it is likely that the minor hypertrophic muscular changes that occur with the RT performed are not sufficient to induce a decrease in mitochondrial density and oxidative potential (15). Also, other factors such as maturation or specific soccer training might have produced this enhancement in endurance performance for U16.

Previous studies (4,15) have also reported a significant relationship between the changes in height jump and acceleration capacity in young soccer players, similar to those found in this study ($r = -0.49$, $p = 0.001$). This relationship indicates that those players with greater improvements in vertical jump likely may obtain greater gains in sprint performance than those with lower increases in jumping ability.

In addition, the relative improvement in CMJ was correlated with the relative increase in V1LOAD ($r = 0.40$). In this case, U16 obtained greater gains in V1LOAD and in CMJ height, whereas U21 obtained a significant improvement in muscle strength without changes in jump performance. Thus, it is likely that the relative increase in leg strength is partly responsible for the improvement in jump height. The fact that U21 improved V1LOAD, and not CMJ height, could suggest that a minimal threshold of strength gains may be necessary to produce these gains in jump performance. However, there was no association between the changes in T20 and V1LOAD ($r = 0.03$). This is in agreement with the study performed by Christou et al. (7), in which a weak correlation was observed between the percent increase in leg strength and 30-m speed ($r = 0.24$). Therefore, these results indicate that the adaptations that produce the improvements in leg strength are different than those that produce the increase in acceleration ability.

Some limitations need to be addressed. The main limitation is that this study is not an experimental design; therefore, it is difficult to obtain a causal relationship between the changes in the different fitness indicators assessed. Other limitation is that we have not measured the player's maturation. Thus, the interpretation of what magnitude of performance improvement is due to maturation or is related to the RT program performed should be interpreted cautiously. Despite these limitations, our results show the importance to add a RT program in young soccer players.

In conclusion, a RT program with moderate loads and low number of repetitions per set combined with jumps and sprints helps foster and accelerate the development of physical performance in young soccer players, as U16 and U18 players equaled or even overcame U21 players in all measured variables of physical performance. In addition, our results also indicate that using movement velocity to monitor and individualize the strength training loads might prevent the use of other methods that cause greater stress, such as the assessment of 1RM or XRM.

Practical Applications

The results of this study can contribute to raise awareness about in-season RT program design for team sports. A practical application for the coach and fitness coach is that RT in this population may be performed with moderate loads (50–65% 1RM), combined with jumps and sprints, because it provides leg strength gains together with noticeable improvements in vertical jump and, to a lesser extent, sprint ability. This suggests that this kind of RT should complement the specific tactical-technical soccer training in the development programs of the professional soccer clubs for increasing the fitness conditioning and to facilitate young soccer players attaining the physical performance necessary to compete with professional soccer players. Furthermore, prescribing and monitoring training load according to movement velocity allows to better control and adjust the programmed load, without the need to perform a 1RM test or a test of maximum number of repetitions to failure.

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